

Statistical Mechanics I Exercise 6

Idan Stark

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1 Landau Theory for Complex Valued Fields

Let us consider the following Landau free energy:

$$\beta\mathcal{F}[\psi(x), \phi(x)] = \int d^d x \left(\frac{K}{2} |\nabla\psi|^2 + t|\psi|^2 + u|\psi|^4 + v|\psi|^6 + \frac{\phi^2}{2\sigma} - \gamma\sigma|\psi|^2 \right) \quad (1)$$

1.1 Effective theory

Let us look at the partition function for that energy:

$$\begin{aligned} Z &= \int D[\psi(x), \phi(x)] \exp[-\beta\mathcal{F}] = \\ &= \int D[\psi(x), \phi(x)] \exp \left[- \int d^d x \left(\frac{K}{2} |\nabla\psi|^2 + t|\psi|^2 + u|\psi|^4 + v|\psi|^6 + \frac{\phi^2}{2\sigma} - \gamma\sigma|\psi|^2 \right) \right] = \\ &= \int D[\psi(x)] \left(\exp \left[- \int d^d x \left(\frac{K}{2} |\nabla\psi|^2 + t|\psi|^2 + u|\psi|^4 + v|\psi|^6 \right) \right] \right) \times \\ &\quad \int D[\phi(x)] \exp \left[- \int d^d x \left(\frac{\phi^2}{2\sigma} - \gamma\sigma|\psi|^2 \right) \right] \end{aligned} \quad (2)$$

The inner integral is the path intergral equivalent of a gaussian, and can be calculated by writing it explicitly and pulling the product out of the integral:

$$\begin{aligned} &\int D[\phi(x)] \exp \left[- \int d^d x \left(\frac{\phi^2}{2\sigma} - \gamma\sigma|\psi|^2 \right) \right] = \\ &= \int \prod_x d\phi(x) \exp \left[- \left(\frac{\phi(x)^2}{2\sigma} - \gamma\phi(x)|\psi|^2 \right) \right] = \\ &= \prod_x \int d\phi(x) \exp \left[- \left(\frac{\phi(x)^2}{2\sigma} - \gamma\phi(x)|\psi|^2 \right) \right] = \\ &= \prod_x e^{\frac{\gamma^2\sigma}{2}|\psi|^4} \int d\phi(x) \exp \left[- \left(\frac{\phi(x)^2}{2\sigma} - \gamma\phi(x)|\psi|^2 + \frac{\gamma^2\sigma}{2}|\psi|^4 \right) \right] = \\ &= \prod_x e^{\frac{\gamma^2\sigma}{2}|\psi|^4} \int d\phi(x) \exp \left[- \frac{(\phi(x) - \gamma\sigma|\psi|^2)^2}{2\sigma} \right] = \\ &= e^{\int d^d x \frac{\gamma^2\sigma}{2}|\psi|^4} \prod_x (\sigma\pi) = \\ &= \exp \left[\int d^d x \left(\frac{\gamma^2\sigma}{2} |\psi|^4 + \ln(\sigma\pi) \right) \right] \end{aligned} \quad (3)$$

This means the integral over $\phi(x)$ added to the effective u in the theory and a constant energy that can be ignored. The effective free energy is then:

$$\beta\mathcal{F}[\psi(x)] = \int d^d x \left(\frac{K}{2} |\nabla\psi|^2 + t |\psi|^2 + \left(u - \frac{\gamma^2\sigma}{2} \right) |\psi|^4 + v |\psi|^6 \right) \quad (4)$$

1.2 Phase Diagram

Let us now use the new energy to find the phase diagram of the effective theory around the minima of the free energy. At that area, using the saddle point approximation, the fluctuations in ψ are much smaller than the field itself ψ . This means we can neglect the first term and focus on the other three. The three remaining terms give the same Landau free energy we saw in assignment 4, which gave a tricritical point due to the ψ^6 term (there is was m^6).

2 Non-Homogeneous Field Model

Some d dimensions system is described by an order parameter that has a special axis, $x_{||}$, namely $m(x_{||}, \vec{x}_{\perp})$. Its Landau-Ginzburg free energy is given by:

$$\beta\mathcal{F}[m(x)] = \int dx_{||} \int dx_{\perp}^{d-1} \left[\frac{t}{2} m^2 + \frac{K}{2} (\partial_{x_{||}} m)^2 + \frac{L}{2} (\nabla_{\perp}^2 m)^2 - hm \right] \quad (5)$$

where $\nabla_{\perp}^2 = \sum_{x_i \neq x_{||}} \partial_{x_i}^2$.

2.1 Symmetries

The first symmetry we notice when $h = 0$ is the $\mathbb{Z}_2$ symmetry $m \rightarrow -m$. The second group of symmetries is the Orthogonal group $O(d-1)$ of rotation and mirroring symmetries for the $d-1$ dimensions $x_i \neq x_{||}$. Lastly, there is a mirroring $\mathbb{Z}_2$ symmetry for the mirroring along $x_{||}$.

2.2 Fourier Space Free energy

Let us now transform the energy to Fourier space. Writing:

$$m = \int \frac{d^d k}{(2\pi)^d} \tilde{m}(\vec{k}) e^{i\vec{k} \cdot \vec{x}} \quad (6)$$

The free energy functional becomes:

$$\begin{aligned} \beta\mathcal{F}[\tilde{m}(\vec{k})] &= \int dx_{||} \int dx_{\perp}^{d-1} \left[\frac{t}{2} m^2 + \frac{K}{2} (\partial_{x_{||}} m)^2 + \frac{L}{2} (\nabla_{\perp}^2 m)^2 - hm \right] = \\ &= \int dx_{||} \int dx_{\perp}^{d-1} \int \frac{d^d k}{(2\pi)^d} \left[\int \frac{d^d k'}{(2\pi)^d} \left(\frac{t}{2} \tilde{m}(\vec{k}) \tilde{m}(\vec{k}') e^{i(\vec{k}+\vec{k}') \cdot \vec{x}} + \right. \right. \\ &\quad \left. \left. + \frac{K}{2} (\partial_{x_{||}} \tilde{m}(\vec{k}) e^{i\vec{k} \cdot \vec{x}}) (\partial_{x_{||}} \tilde{m}(\vec{k}') e^{i\vec{k}' \cdot \vec{x}}) + \frac{L}{2} (\nabla_{\perp}^2 \tilde{m}(\vec{k}) \tilde{m}(\vec{k}) e^{i\vec{k} \cdot \vec{x}}) (\nabla_{\perp}^2 \tilde{m}(\vec{k}') \tilde{m}(\vec{k}') e^{i\vec{k}' \cdot \vec{x}}) \right) - h \tilde{m}(\vec{k}) e^{i\vec{k} \cdot \vec{x}} \right] \end{aligned} \quad (7)$$

Each of the terms where there are two $\tilde{m}$, evaluating the space integral over the exponents gives a delta function $(2\pi)^d \delta(\vec{k} + \vec{k}')$, while last term that has only one exponent gives $(2\pi)^d \delta(\vec{k})$. So, using these delta functions to integrate over the relevant k integrals gives:

$$\beta\mathcal{F}[\tilde{m}(k_{||}, \vec{k}_{\perp})] = \int \frac{d^d k}{(2\pi)^d} \tilde{m}(k_{||}, \vec{k}_{\perp}) \tilde{m}(-k_{||}, -\vec{k}_{\perp}) \left(\frac{t}{2} + \frac{K}{2} k_{||}^2 + \frac{L}{2} \vec{k}_{\perp}^4 \right) - h \tilde{m}(0, \vec{0}) \quad (8)$$

2.3 RG Recursion relations

Let us now find the recursion relations for the constants in the theory. For this, let us rescale $k'_{||} = bk_{||}$ and $\vec{k}'_{\perp} = c\vec{k}_{\perp}$ and renormalize the field $\tilde{m}' = m/z$. This gives:

$$\beta\mathcal{F}[\tilde{m}(k'_{||}, \vec{k}'_{\perp})] = \int \frac{d^d k'}{(2\pi)^d} \frac{z^2}{bc^{d-1}} \tilde{m}(k'_{||}, \vec{k}'_{\perp}) \tilde{m}(-k'_{||}, -\vec{k}'_{\perp}) \left(\frac{t}{2} + \frac{K}{2b^2} k'^2_{||} + \frac{L}{2c^4} \vec{k}'^4_{\perp} \right) - h\tilde{m}'(0, \vec{0}) \quad (9)$$

This gives the recursion relations:

$$\begin{aligned} t'/t &= z^2 b^{-1} c^{1-d} \\ K'/K &= z^2 b^{-3} c^{1-d} \\ L'/L &= z^2 b^{-1} c^{-3-d} \end{aligned} \quad (10)$$

2.4 Requiring unchanged prefactors

Next, let us require that $K' = K$ and $L' = L$. This gives:

$$\begin{cases} z^2 b^{-3} c^{1-d} = 1 \\ z^2 b^{-1} c^{-3-d} = 1 \end{cases} \quad (11)$$

Solving the equations for $c(b)$ and $z(b)$ gives:

$$c(b) = b^{1/2} \quad z = b^{(d+5)/4} \quad (12)$$

This means that:

$$t' = b^2 t \quad (13)$$

2.5 Critical Exponents

Let us finally find the critical exponent ν , for the correlation length:

$$\xi(t) \sim t^{-\nu} \quad (14)$$

At ever RG space, the correlation length scales according to the constants b and c :

$$\xi'_{||}(t') = b^{-1} \xi_{||}(t) \quad \xi'_{\perp}(t') = c^{-1} \xi(t) = b^{-1/2} \xi_{\perp}(t) \quad (15)$$

So:

$$t'^{-\nu_{||}} = b^{-2\nu_{||}} t^{-\nu_{||}} \sim \xi'_{||}(t') = b^{-1} \xi_{||}(t) \sim b^{-1} t^{-\nu_{||}} \rightarrow \nu_{||} = \frac{1}{2} \quad (16)$$

$$t'^{-\nu_{\perp}} = b^{-2\nu_{\perp}} t^{-\nu_{\perp}} \sim \xi'_{\perp}(t') = b^{-1/2} \xi_{\perp}(t) \sim b^{-1/2} t^{-\nu_{\perp}} \rightarrow \nu_{\perp} = \frac{1}{4} \quad (17)$$

3 Upper critical dimension

Let us consider the Landua-Ginzburg free energy:

$$\mathcal{F}[m(x)] = \frac{1}{2} (\nabla m)^2 + tm^2 + um^4 \quad (18)$$

In class, we saw that for every natural $n > 2$, the prefactor g_n of m^n changes under RG flow to be:

$$g'_n = g_n l^{-dn/2+n+d} \quad (19)$$

Applying this rule for $u = g_4$, we get:

$$u(l) \sim l^{4-d} \quad (20)$$

Let us take $d = 5$, since it's the lowest dimension above the UCD. This gives:

$$u(l) \sim l^{-1} \quad (21)$$

Therefore, for $l_0 = 100$, we get a $u(l_0) \ll 1$. We can now solve the ODE for t , which is the regular gaussian term:

$$\frac{dt}{dl} = 2lt \quad (22)$$

Which gives:

$$t = t_0 e^{l^2} \quad (23)$$

And, plugging $l_0 = 100$ we get:

$$t = e^{10000} t_0 \quad (24)$$

Which is much larger than the $u(l_0)$ term. This demonstrates that under RG flow for $d > 4$, u grows vanishingly small with respect to the other terms. Since we took the lowest dimension, this will also be true for any dimension above the UCD $d > 4$.

The last term, in k -space, becomes:

$$\int d^d r u m^4(r) = \frac{u}{(2\pi)^{4d}} \int d^d k_1 \cdots d^d k_4 \tilde{m}(k_1) \cdots \tilde{m}(k_4) (2\pi)^d \delta(k_1 + k_2 + k_3 + k_4) \quad (25)$$

This, unlike the m^2 term, couples the $k = 0$ terms with the $k \neq 0$ terms. When u is small, this coupling can be neglected, and we can write a theory for $k = 0$ alone. It will include the two original terms for $k = 0$, and the term in the above integral, with all four momenta being zero. This gives:

$$\mathcal{F}[\tilde{m}(0)] = \frac{1}{2} t \tilde{m}^2(0) + u \tilde{m}^4(0) \quad (26)$$

This theory for $k = 0$ takes the same structure as a Landau theory, and so would give the same critical exponents as the Landau theory.

Now let us consider the following Landau-Ginzburg free energy:

$$\mathcal{F}[m(x)] = \frac{1}{2} (\nabla m)^2 + t m^2 + v m^6 \quad (27)$$

and find its UCD using the different approaches.

3.1 Ginzburg argument

Let us calculate the Ginzburg ratio:

$$R = \frac{\int_{|\mathbf{r}| < \xi} d^d r \langle m(\mathbf{r})^2 \rangle}{\xi^d m_0^2} \quad (28)$$

First, we start by calculating the nominator, which is the fluctuations of m . We go above the transition where $m = 0$, allowing us to ignore the m^6 term, and get the same gaussian approximation we got for the m^4 power in class. This gives the same nominator:

$$\int_{|\mathbf{r}| < \xi} d^d r \langle m(\mathbf{r})^2 \rangle \sim t^{-1} \quad (29)$$

The denominator, however, is different. Let us minimize the free energy as a function of m to find m_0 :

$$\frac{dF}{dm} = 2mt + 6vm^5 \rightarrow m \sim t^{1/4} \quad (30)$$

So, the denominator is:

$$\xi^d m_0^2 \sim t^{\frac{1-d}{2}} \quad (31)$$

$$R \sim \frac{t^{-1}}{t^{\frac{1-d}{2}}} = t^{\frac{1}{2}(d-6)} \quad (32)$$

Which gives the UCD $d = 6$.

3.2 K-space RG

Let us now find the same from K-space RG to first order. The K-space free energy is:

$$\mathcal{F}[\tilde{m}(k)] = \int^\Lambda \frac{d^d k}{(2\pi)^d} \frac{1}{2} (k^2 + t) \tilde{m}(k) \tilde{m}(-k) + \frac{v}{(2\pi)^{5d}} \int^\Lambda d^d k_1 \cdots d^d k_6 \tilde{m}(k_1) \cdots \tilde{m}(k_6) \delta(k_1 + \cdots + k_6) \quad (33)$$

This gives the same partition function as the m^4 term we saw in class:

$$Z = \int \mathcal{D}[\tilde{m}^-] \left[e^{-\frac{1}{2} \int_0^{\Lambda/b} \frac{d^d k}{(2\pi)^d} (t+k^2) \tilde{m}^-(k) \tilde{m}^+(-k)} \int \mathcal{D}[\tilde{m}^+] e^{-\frac{1}{2} \int_{\Lambda/b}^\Lambda \frac{d^d k}{(2\pi)^d} (t+k^2) \tilde{m}^-(k) \tilde{m}^+(-k)} e^{-F_I(\tilde{m}^+ \tilde{m}^-)} \right] \quad (34)$$

Only with the interaction:

$$F_I(\tilde{m}^+, \tilde{m}^-) = \frac{v}{(2\pi)^{5d}} \int_0^\Lambda d^d k_1 \cdots d^d k_6 \tilde{m}(k_1) \cdots \tilde{m}(k_6) \delta(k_1 + \cdots + k_6) \quad (35)$$

The inner integral is again the same as doing a distribution

$$\int \mathcal{D}[\tilde{m}^+] e^{-\frac{1}{2} \int_{\Lambda/b}^\Lambda \frac{d^d k}{(2\pi)^d} (t+k^2) \tilde{m}^-(k) \tilde{m}^+(-k)} e^{-F_I(\tilde{m}^+ \tilde{m}^-)} = Z_{m^+} \langle e^{-F_I} \rangle \quad (36)$$

And to first order:

$$\langle e^{-F_I} \rangle = 1 - \langle F_I \rangle \quad (37)$$

So, evaluating $\langle F_I \rangle$, we get three surviving terms - the terms with two, four, and six m^+ :

$$\tilde{m}_1^- \tilde{m}_2^- \tilde{m}_3^- \tilde{m}_4^- \langle \tilde{m}_5^+ \tilde{m}_6^+ \rangle \quad \tilde{m}_1^- \tilde{m}_2^- \langle \tilde{m}_3^+ \tilde{m}_4^+ \tilde{m}_5^+ \tilde{m}_6^+ \rangle \quad \langle \tilde{m}_1^+ \tilde{m}_2^+ \tilde{m}_3^+ \tilde{m}_4^+ \tilde{m}_5^+ \tilde{m}_6^+ \rangle \quad (38)$$

The last term is just a constant, and so we focus on the first two terms. The first term gives the gaussian integral:

$$\langle \tilde{m}_5^+ \tilde{m}_6^+ \rangle = \frac{(2\pi)^d \delta(k_5 + k_6)}{t + k_5^2} \quad (39)$$

The delta function gives $k_5 = -k_6$, which together with the original delta function also gives $k_1 = -k_2$. Plugging this into the integral over $\tilde{m}_1^- \tilde{m}_2^-$, we get the first term contribution:

$$6v \left(\int \frac{d^d k}{(2\pi)^d} \tilde{m}^-(k) \tilde{m}^+(-k) \right) \left(\int \frac{d^d q}{(2\pi)^d} \frac{1}{t + q^2} \right) \quad (40)$$

The second term, however, is a bit more complex, since it has an average over four m^+ . This means we have to split it using Wick's theorem. We did it in class, and the result was the same as the two m^+ , only with the denominator squared and the prefactor 36:

$$36v \left(\int \frac{d^d k}{(2\pi)^d} \tilde{m}^-(k) \tilde{m}^+(-k) \right) \left(\int \frac{d^d q}{(2\pi)^d} \frac{1}{(t + q^2)^2} \right) \quad (41)$$

And so, up to a constant:

$$\langle F_I \rangle = \left(\int \frac{d^d k}{(2\pi)^d} \tilde{m}^-(k) \tilde{m}^-(-k) \right) \left(6 \int \frac{d^d q}{(2\pi)^d} \frac{1}{t + q^2} + 36 \int \frac{d^d q}{(2\pi)^d} \frac{1}{(t + q^2)^2} \right) \quad (42)$$

And so, the flow of t is now given by:

$$\bar{t} = t + \int_{\Lambda/b}^{\Lambda} \frac{d^d q}{(2\pi)^d} \frac{12}{t + q^2} + \frac{72}{(t + q^2)^2} \quad (43)$$

Let us perform the rescaling and renormalization steps:

$$k' = bk \quad \tilde{m}' = b^\omega \tilde{m} \quad (44)$$

This gives the free energy:

$$\begin{aligned} \mathcal{F}[\tilde{m}(k')] &= \frac{1}{2} \int_0^{\Lambda/b} \frac{d^d k'}{(2\pi)^d} b^{-d} (\bar{t} + b^{-2} k'^2) b^{-2\omega} \tilde{m}'(k) \tilde{m}'(-k) + \\ &+ v \int_0^{\Lambda/b} \frac{d^d k'_1 \dots d^d k'_6}{(2\pi)^{5d}} b^{-5d} b^{-6\omega} \tilde{m}'(k_1) \dots \tilde{m}'(k_6) \end{aligned} \quad (45)$$

Again the k^2 coefficient must be 1:

$$b^{-2d-2-2\omega} = 1 \rightarrow \omega = -\frac{d}{2} - 1 \quad (46)$$

Which gives the recursion relations:

$$\begin{aligned} t' &= b^2 \left(t + \int_{\Lambda/b}^{\Lambda} \frac{d^d q}{(2\pi)^d} \frac{12}{t + q^2} + \frac{72}{(t + q^2)^2} \right) \\ v' &= b^{-5d-6\omega} v = b^{6-d} v \end{aligned} \quad (47)$$

This gives the UCD $d = 6$.